

AI-Assisted Development Workflow

1. Overview

This project was built using a hybrid AI-assisted approach leveraging GitHub Copilot and ChatGPT and other tools.

AI was used as a **coding accelerator and design assistant**, while all key decisions, validations, and architecture were driven through structured reasoning and manual verification.

2. How AI Tools Were Used

Dataset Understanding

- Used AI prompts to analyze multiple JSONL files
- Identified entities, keys, and cross-file relationships
- Mapped the core ERP flow:
 - Order → Delivery → Invoice → Journal → Payment

Graph Design

- Used AI to propose graph modeling (nodes + edges)
- Iteratively refined to **item-level relationships**
- Final model captured real ERP behavior instead of simplistic joins

Backend & Query Engine

- Generated FastAPI structure using AI
- Built graph traversal functions with AI assistance:

- trace order flow
- find journal entries
- detect missing invoices
- Validated logic manually to ensure correctness

LLM Integration (Gemini)

- Used AI to design structured prompt templates
- LLM used only for:
 - intent extraction
- All data answers come from graph traversal (no hallucination risk)

Frontend

- Generated minimal React chat UI via AI
- Focused on functionality over design for fast iteration

3. Prompting Strategy

- Broke system into modules (data → graph → API → LLM → UI)
- Used **explicit, structured prompts** with clear outputs
- Iteratively refined prompts based on results
- Enforced:
 - JSON-only outputs
 - strict intent formats

4. Debugging & Iteration

Key issues identified and resolved:

- **Data inconsistencies**
 - Item IDs (`000010` vs `10`) → normalized
- **Relationship accuracy**
 - Not all links are strict FKs → modeled optional relationships
- **System validation**
 - Tested via FastAPI `/docs`
 - Verified multi-step flows end-to-end
- **LLM reliability**
 - Added fallback heuristic parser
 - Ensured system works without LLM dependency

5. Guardrails

- Strict intent schema
- Unknown query handling

- LLM limited to parsing only
- Deterministic execution via graph

6. Key Learnings

- ERP systems require strong data modeling before building logic
- Graph structures are ideal for process tracing
- LLMs are best used for **interpretation, not computation**
- Combining AI + deterministic logic improves reliability
- Iterative refinement is critical for correctness

7. Summary

AI significantly accelerated development by enabling:

- rapid prototyping
- structured exploration
- modular implementation

Final system correctness was ensured through:

- validation
- controlled use of AI
- clear architectural decisions

This hybrid approach enabled building a robust, end-to-end solution efficiently.